

Primal Dual — Finiteness of Algorithm

Recall the basic setting.

Idea: use C.S. — if $\exists x$ which satisfies
CS w/ dual ~~was~~ feasible p , then (x, p) opt.
o/w: improve p .

<u>P</u>	<u>D</u>
min: $c^T x$	max: $p^T b$
st: $Ax = b$	st: $p^T A \leq c$
$x \geq 0$	p free

wlog $b \geq 0$.

Dual feasible p
 $J = \{j: p^T A_j = c_j\}$

RP:

$$\begin{aligned} \min: & \sum x_i^a \\ \text{st: } & A_j x_j + x^a = b \\ & x_j^a = 0 \\ & x^a \geq 0 \\ & x_j \geq 0 \end{aligned}$$

DRP:

$$\begin{aligned} \max: & p^T b \\ \text{st: } & p^T A_j \leq c_j \quad j \in J \quad \text{c is given} \\ & p \leq 1 \\ & p \text{ free} \end{aligned}$$

$\bar{p}$ s.t. a b DRP:

$p_{\text{new}} = p + \theta \bar{p}$, θ determined by

$$\min_{j \in J^c} \frac{c_j - p^T A_j}{p^T A_j}$$

Properties of the P-D Algorithm

Thm: If $\underline{RP} > 0$, and if $\bar{p}^T A_j \leq 0 \quad \forall j$
 $\Rightarrow$ dual is unbd $\Rightarrow$ primal infeasible.

Thm: Let B be the ~~best~~ opt basis for the RP. $B \subseteq J \cup [m]$. Then: $B \cap J \subseteq J_{\text{new}}$
ie., if $j \in B \cap J$ ($\bar{p}_j^T A_j = c_j$), then

$$(\bar{p}_j + \theta \bar{p})^T A_j = c_j, \text{ hence } j \in J_{\text{new}}.$$

pf: $\bar{p}^T A_j \leq 0$ b/c of constraints, but,

if $j \in B$, its reduced cost in RP is zero

$$\underline{\text{In RP,}} \quad \bar{c}_j = 0 - \boxed{\bar{p}^T A_j = 0} \quad \square$$

Hence, we can start simplex method at previous sol'n.

Finiteness of P-D

Let p be dual sol'n. Let $\bar{p}$ be sol'n to DRP.

Suppose $\bar{p}^T b > 0$. (else optimal, & stop).

assume $\exists j \in J^C$ w/ $\bar{p}^T A_j > 0$. o.w., (P) inf. & stop.

$$\theta = \min_{\substack{j \in J^C \\ \bar{p}^T A_j > 0}} \frac{c_j - \bar{p}^T A_j}{\bar{p}^T A_j} \quad // \quad j_0.$$

$$\text{choice of } \theta: \quad p_{\text{new}}^T A_{j_0} = (p + \theta \bar{p})^T A_{j_0} = c_{j_0} \Rightarrow \underline{j_0 \in J_{\text{new}}}.$$

Next iteration of RP: Reduced cost of j_0 : $\bar{c}_{j_0} = 0 - \bar{p}^T A_{j_0} < 0$

$\Rightarrow$ we pivot.

If no degeneracy $\Rightarrow$ value of RP strictly reduces $\Rightarrow$ finite termination.

If we have degeneracy, anticycling rule + finite # of BFs $\nearrow$.

Coming up

PD - "combinatorializes" our LP.